

CS 255 System Design Document

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UML Diagrams

UML Use Case Diagram...	... See Figure 1 (pg. 3)
UML Activity Diagrams...	... See Figure 2-1,2-2 (pg. 3)
UML Sequence Diagram...	... See Figure 3 (pg. 4)
UML Class Diagram...	... See Figure 4 (pg. 5)

Technical Requirements

Hardware Requirements

- Client devices: Standard desktop computers, tablets, and smartphones with modern web browsers
- No specialized hardware required

Software Requirements

- Web-based application accessible through standard web browsers
- Browser compatibility: Chrome, Firefox, Safari, Edge (current versions)
- Responsive design for mobile and desktop viewing
- No platform-specific applications required

Database Requirements

- Relational database to store:
 - o Customer account information
 - o Training package enrollment data
 - o Driving lesson schedules and reservations
 - o Online test progress and scores
 - o Driver and vehicle assignments
 - o Audit logs of all user actions

Platform & Infrastructure Requirements

- Cloud-hosted web application (AWS, Azure, or Google Cloud)
- Cloud provider manages:
 - o Data backups
 - o Security maintenance
 - o System uptime and availability
 - o Data encryption
- Web server and application server for hosting the system

API & Integration Requirements

- Secure HTTPS encryption for all data transmission
- Payment processing integration for credit card transactions
- DMV system integration for receiving rule and policy updates
- Email notification system for confirmations

Authentication & Security

- Username and password login system
- Password reset capability for forgotten passwords
- Role-based access control for four user types (Customer, Secretary, Admin, Owner)
- Account lockout after multiple failed login attempts
- Encrypted storage of sensitive data (credit card, personal information)

Performance Requirements

- System available during business hours (8 AM - 6 PM)
- Fast response times for user actions
- Support multiple concurrent users without slowdowns
- Automatic driver/vehicle assignment without delays

Figure 1: UML Case Diagram

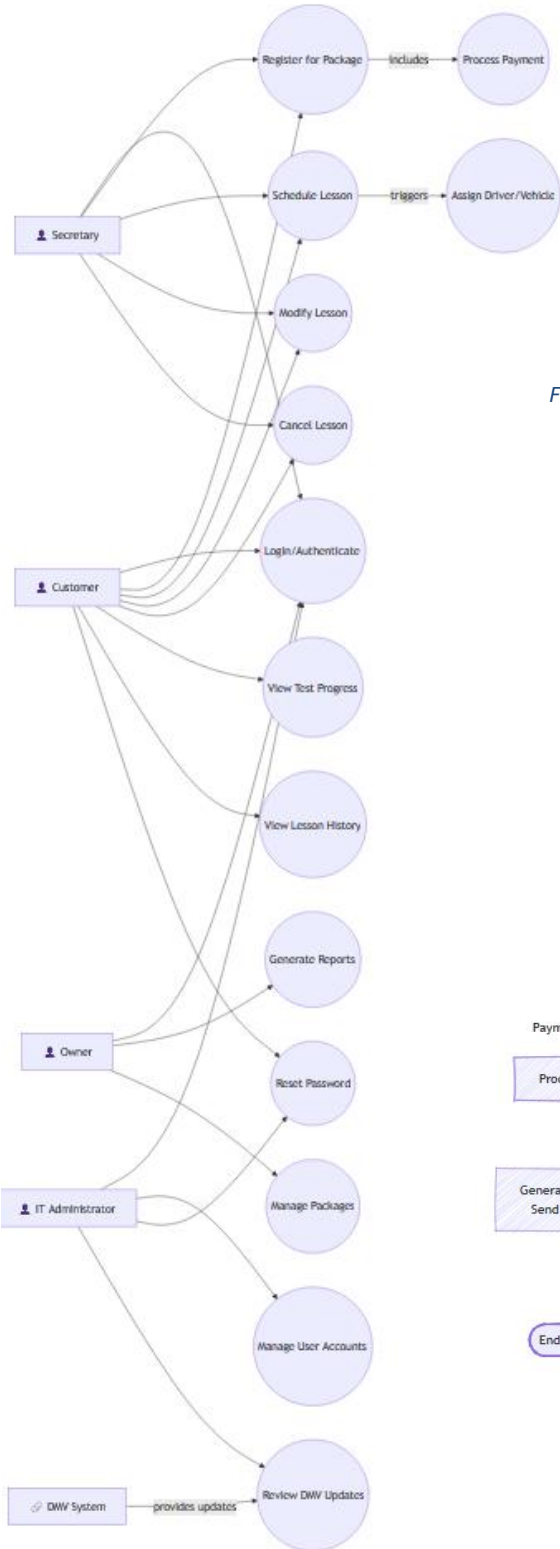


Figure 2-1: Activity - Register for Package

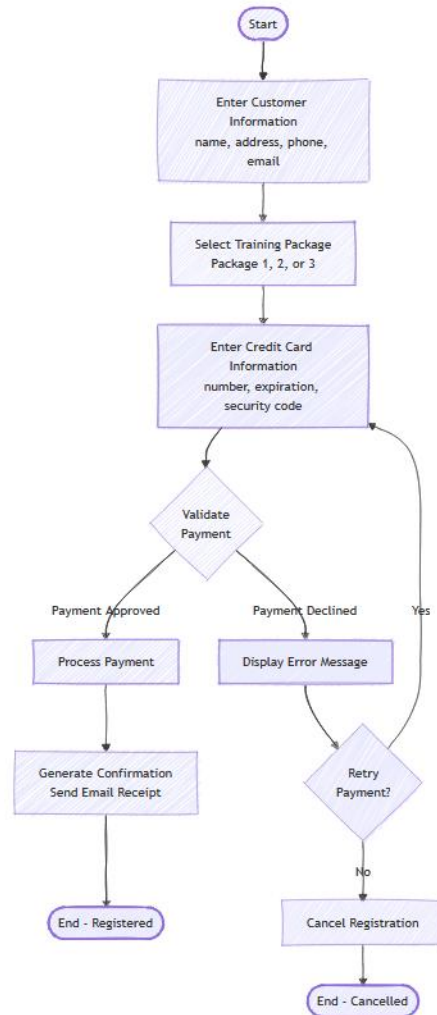


Figure 2-2: Activity - Schedule Lesson

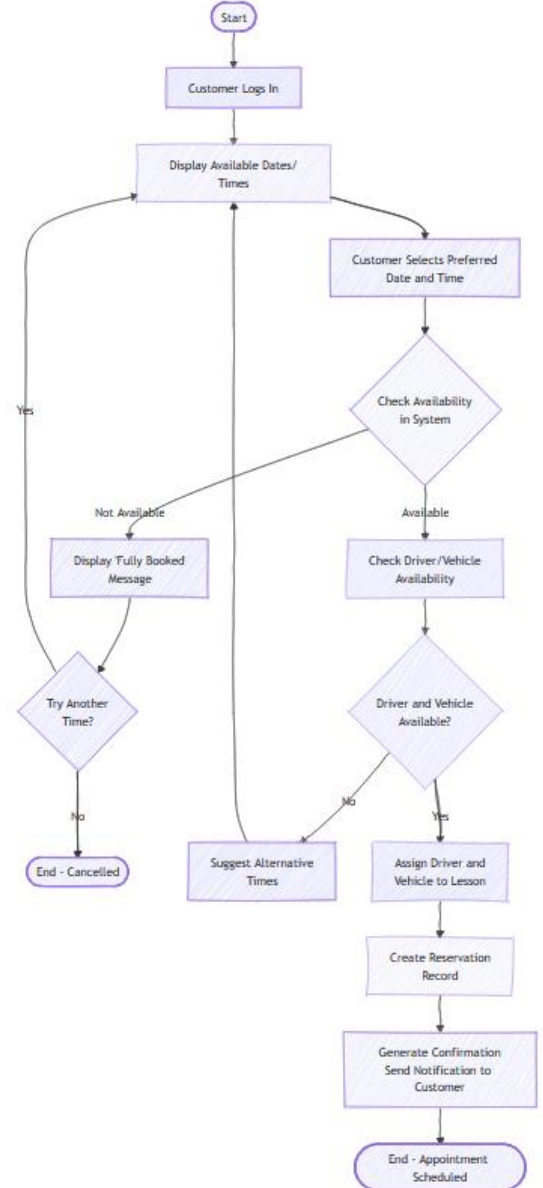


Figure 3: Sequence Diagram

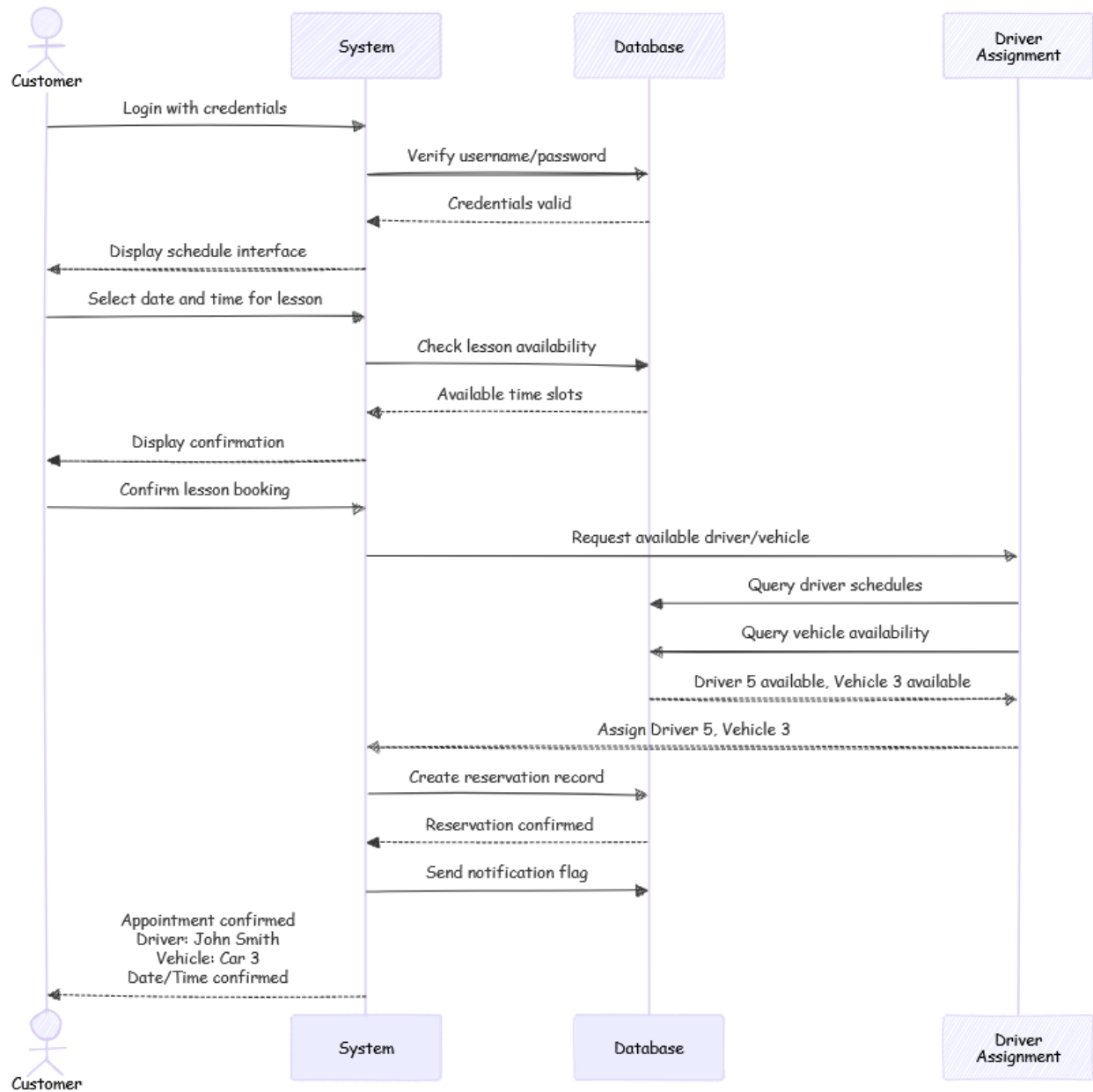


Figure 4: Class Diagram

